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| **Status** | Finished |
| **Started** | Tuesday, 20 October 2026, 7:34 PM |
| **Completed** | Tuesday, 20 October 2026, 7:40 PM |
| **Duration** | 6 mins 6 secs |
| **Grade** | **12.00** out of 15.00 (**80**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 89 and 94
- c. 94 and 99
- d. 90 and 100

Question **2**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. 1
- b. n/2
- c. n ✓
- d. log n

Question **3**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Divide the list into two equal parts
- b. Sort the list ✗
- c. Compare the target element with the middle element in the list
- d. Compare the target element with the first element in the list

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By starting from the first element
c. By dividing the list length by two ✓
d. By comparing each element sequentially

Question **5**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n) ✓
d. O👎

Question **6**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99 ✗
b. 90 and 99
c. 90 and 94
d. 89 and 94

Question **7**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Binary ✓

c. Linear

d. Hash

Question **8**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2

b. log n

c. n-1

d. n ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(1) ✓

b. O(log n)

c. O(n log n)

d. O👎

Question **10**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element

b. The middle element is chosen randomly

c. The search stops

d. The lower middle element is chosen as the middle element ✗

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist

b. The list must be sorted

c. Number values should only be present

Number values should only be present

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **12**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list

b. In the right sublist ✓

c. In the left sublist

d. In the middle of the list

Question **13**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Determining whether an element is present in a list ✓

b. Inserting elements into a list

c. Deleting elements from a list

d. Sorting elements in a list

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O(n log n)
c. O👎 ✓
d. O(log n)

Question **15**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 10 ✓
c. 500
d. 1000